Surfacing Business Associates from Public Registry Data: Embedding-Based Candidate Retrieval over Japan's Corporate Registry

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Abstract

Identifying relationships between companies is a core requirement of due diligence and anti money-laundering practices, but in Japan, the public corporate registry data records no ownership information and has limited representative data in open bulk sources. This study asks whether two fields that are available, company name and registered address, can serve as proxy signals to find out to what extent does semantic similarity over these fields surface structural association between companies, and how much of what is surfaced is just a resemblance. Each of 1,173,931 active Tokyo legal persons from the government's gBizINFO dataset is represented as a single normalized name and address fingerprint, embedded with a 300 million parameter multilingual text embedding model, and searched against all others by approximate nearest neighbor retrieval. The retrieved candidate pairs above a cosine similarity threshold are merged into 39,563 candidate clusters by connected components and evaluated through structured manual review. The results revealed that there is an asymmetry of relatedness that is inscribed formulaically and method's failures. Formulaic relatedness in a form of duplicate records, paired securitisation structures, and administration hubs is retrieved with high-confidence, while organically expressed group structures are invisible by the embedding representation. The method's failures arise where common vocabulary and shared premises imitate the formulaic pattern through transitive chaining. The study thereby measures the practical ceiling of Japan's open registry data for network analysis of what public fields expose the corporate-vehicle layer of the economy which are well-described in beneficial ownership literature as instruments of concealment, and only that part of it which entities choose to broadcast.

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Introduction

Corporate ownership is meant to be legible on paper, but is routinely illegible in practice. A World Bank review of 150 grand corruption cases found that concealing the beneficial owner through corporate vehicles was nearly universal (Van der Does de Willebois et al., 2011), and the international response to this problem, led by the Financial Action Task Force, has centered on requiring countries to make beneficial ownership information available to competent authorities (FATF, 2023). Following the 2022 revisions of FATF's framework and the October 2024 Follow-Up Report, Japan no longer holds any recommendations "partially compliant" (FATF, 2024). Yet this compliance is addressed to competent authorities, rather than the public. As of now, Japan maintains no publicly searchable register of beneficial owners, the 2022 Beneficial Ownership List System is voluntary and closed to public search (Ministry of Justice, 2023). The practical route to ownership and relationship data runs through commercial providers such as Tokyo Shoko Research, Teikoku Databank, or Orbis. The sizable information density gap between public and commercial datasets raises the question of whether the public dataset can serve as a substitute for something like business associates even after the 2024 reforms recognized by FATF. For a compliance officer, journalist, or a student working from open data, the gap between compliance on paper and access in practice is the operative fact.

Nevertheless, Japan does publish a corporate registry without restriction. The National Tax Agency assigns every legal person a corporate number and publishes its registered name and headquarters address. The Ministry of Economy, Trade, and Industry gBizINFO's service aggregates it into a downloadable file called 基本情報, or Kihonjoho, of roughly 5.8 million records (Ministry of Economy, Trade and Industry [METI], n.d.). This data contains 96 columns, but no shareholding fields and its only representative field is filled for around 1.3% of Tokyo records. On the surface, this dataset does not provide an answer to who owns what. The central question of this thesis is whether this dataset can answer something adjacent, but still useful: which companies are associated with each other. Although, business associate relationships such as shared administration, succession, common sponsorship, are not ownership links, but they are the surrounding network in which informal control leaves traces (Vitali, Glattfelder & Battiston, 2011), and in a registry with limited available information, they are the only relationships that could plausibly be recovered at all. Rather than resolving the beneficial owner of a single entity, the present work seeks to surface business associates which are distinct companies that are structurally connected through shared registry attributes like a common address, representative, or name element, in a way that ownership data alone struggles to reveal.

The signals for such recovery are established from prior work where shared addresses and shared officers formed meaningful edges in networks built from registry and data leaks

(Joaristi, Serra & Spezzano, 2019; Kejriwal & Dang, 2020), and name similarity is a long-standing cue of relatedness in entity resolution (Christen, 2012). The method's question is how to use these signals at the registry scale with millions of entities. Although classical string distance is effective against typographic variation, it is blind to relatedness that is not "spelled out" (Cohen, Ravikumar & Fienberg, 2003), which motivated the choice of learned text embeddings, where similarity is measured as proximity in a vector space (Reimers & Gurevych, 2019). However, embeddings carry a risk, which is documented by Mtumbuka and Schockaert (2024), that representations built from entity attributes tend to capture what entities are rather than how they are related. Two companies in the same office building or two founders independently choosing the same English word for a name are similar without being structurally connected. Therefore, this thesis is set to find out how much of embedding similarity over public registry attributes reflects structural association between Japanese companies. It will also explore what type of structures does the method surface, where does the usable similarity threshold lie and what are the limits of Japan's open registry data placed on associate identification.

Literature Review

UBO Resolution & Corporate Due Diligence

UBO resolution is important for due diligence, but why is it important in Japan's case and are proxy signals from public data necessary?

Due diligence, a process of enquiry and investigation made by a prospective purchaser in order to confirm the authenticity of its purchase, is a crucial practice in financial sectors and beyond in all countries around the world (Howson, 2017). Due diligence is simultaneously a form of risk mitigation, value maximization, and verification process across multiple forms of business transactions, including firm mergers, real estate purchases and corporate investments. Within government entities, due diligence plays a role in supporting anti-money laundering (AML) and counter-terrorist financing (CFT) operations. This is the primary role of the Financial Action Task Force (FATF), a 40-member inter-governmental body, founded by G7 countries to guide global response to money laundering, terrorist and proliferation financing. (FATF, n.d.). FATF operates by researching how money is laundered and terrorism is financed, setting international standards to reduce those risks, and evaluating whether nations are acting effectively.

Their main instrument to ensure those goals is its framework of 40 Recommendations. First drawn up in 1990 to combat the laundering of drug money through the financial system, the

Recommendations have been revised repeatedly to reflect evolving techniques to commit financial crime. They are recognized as the international standard for AML/CFT, which countries are expected to implement through measures adapted to their particular circumstances (FATF, 2012). The Recommendations divide into 7 distinct areas: AML/CFT policies and coordination, money laundering and confiscation, terrorist financing and financing of proliferation,preventive measures, transparency and beneficial ownership of legal persons and arrangements, powers and responsibilities of competent authorities and other institutional measures, and international cooperation (FATF, 2012).

Among these areas, a topic central to this thesis is transparency and beneficial ownership of legal persons, governed by Recommendation 24 (R24). This topic is deeply explored by FATF because ownership information can be obscured and conceal, in many respects, the real reason for holding assets and conducting transactions (FATF, 2023). They regularly update and publish reports, recommendations, and per-country evaluations of how well those standards are maintained. In 2022, FATF revised R24, strengthening the standard by requiring countries to ensure that competent authorities have access to timely and accurate beneficial ownership information, and by proposing a multi-pronged approach where, instead of relying on companies' self-reporting, countries must use several mechanisms to collect beneficial ownership data. A 2023 publication, entitled "Guidance on Beneficial Ownership of Legal Persons," explains how countries are expected to implement these revised requirements (FATF, 2023). The counter actions from FATF are justified by the scale of this problem worldwide: a World Bank review of 150 grand corruption cases found that concealing the beneficial owner through corporate vehicles was nearly universal (Van der Does de Willebois et al., 2011).

The same review by van der Does de Willebois also argues that beneficial ownership is best understood as a substantive matter of de facto control and benefit derived, not a legal formality like a directorship or a fixed shareholding threshold which means that, at best, formal screening leads to suggestive inferences and "people of interest" rather than identify real controller (Van der Does de Willebois et al., 2011). Moreover, in a field experiment in which the authors posed as consultants and made thousands of incorporation requests to corporate service providers worldwide, wealthy developed countries were often the worst at enforcing basic identity checks (Findley, Nielson & Sharman, 2014). While this concern usually centers on identifying the controlling individual behind a company, the same opacity hides the relationships between companies. Therefore, identifying beneficial owners cannot be separated from identifying their associates since the informal control leaves only traces in the surrounding network rather than in any single ownership field. In Japan specifically, where public registry data oftentimes does not record beneficial ownership information, these associate relationships must be derived from circumstantial information, like names, addresses, representatives or other

substantial signals; private companies, like Tokyo Shoko Research, may also maintain databases on these kinds of corporate relationships. (TSR, n.d.).

These signals matter in ultimate beneficial owner (UBO) resolution because corporate control can be distributed through indirect and circular ownership chains; control over an entity may not be held by the registered shareholder but could instead exist several layers removed in the ownership network (Vitali, Glattfelder & Battiston, 2011). While this information is not a replacement for ownership links, these relationships are overlooked by standard screening and ownership-based resolution technology. Corporate control can be exercised at a distance through intermediate holdings, meaning that relationships that link entities become indicators that can potentially identify the actual controllers, motivating a network-based approach to this problem. Nevertheless, associate identification should not be viewed as a substitute for crucial ownership data, but rather a response to its limits.

Japan illustrates the gap between standards on paper and access in practice. As a G7 member, Japan is a founding participant in FATF and is subject to its per-country Mutual Evaluation Reports (MER). The 2021 MER credited Japan with robust action in areas such as proliferation financing, virtual assets, and AML/CFT, but recommended stronger customer due diligence requirements for both financial and non-financial institutions as well as improved access to beneficial ownership information (Financial Action Task Force [FATF], June 2021). The October 2024 Follow-Up Report noted further progress by re-rating Japan in Recommendation 24 (R24), “Transparency and beneficial ownership of legal persons;” as of that report, Japan no longer holds any recommendation rated “partially compliant” (FATF, 2024). However, for anyone sourcing information from public data, this progress is less meaningful than it appears. R24 only requires that competent authorities such as law enforcement and the financial intelligence unit have access to beneficial ownership information, not the general public. A strong compliance rating does not necessarily mean that a company or a researcher working from open data can reach it. In January 2022, Japan introduced the Beneficial Ownership of Legal Persons List System, administered through the commercial registry by the Legal Affairs Bureau, as a mechanism to record beneficial owners of a company's establishment (Ministry of Justice, 2023). Under this system, a beneficial owner is a natural person holding more than one-fourth of a company’s voting rights, following the definition in the Act on Prevention of Transfer of Criminal Proceeds. However, since the registration is voluntary and the resulting list is not publicly searchable, Japan does not currently maintain a public UBO register for companies.

In the absence of a reliable public register, the practical route to this information is commercial. To purchase corporate data from credit-research agencies, compliance teams can use

domestic providers such as Tokyo Shoko Research (TSR) or Teikoku Databank (TDB), or international providers like Moody's Orbis (TSR, n.d.; TDB, n.d.; Moody's, n.d). Even foundational academic work in corporate control networks has relied on commercial databases such as Orbis (Vitali et al., 2011). The aforementioned commercial datasets - particularly TSR - include information as granular as company suppliers, manager names, type of shareholders, and CEO's birthdates. The sizable information density gap between public and commercial datasets raises the question of whether the public dataset can serve as a substitute for something like business associates even after the 2024 reforms recognized by FATF.

Network Inference from Public Registry Data

Are shared address, shared director, and name similarity across registry records meaningful signals for inferring that companies are operationally or structurally related, even without ownership data?

Record linkage, more formally referred to as entity resolution, is the process by which an investigator determines whether various records in public registry data are associated with a specific firm or more than one; in the latter case, the investigator may infer relationships between the companies (Fellegi & Sunter, 1969). Direct comparison of datasets is not necessarily straightforward, as they may differ based on their organization, structure, or means of identification; however, significant work has been done on standardizing dataset fields, comparison methods, and classification of results. The use of similarity measurements between short-text values, such as names and addresses, have also been introduced (Christen, 2012). Most of this toolkit, including edit-distance and token-based string-similarity measures, was developed to facilitate deduplication, a process of identifying records that refer to the same real-world entity. In this study, exact-entity deduplication is unnecessary, since every company has a unique identifier by its corporate number (法人番号). Instead of matching, the study is aimed at testing embeddings as a method for surfacing related entities.

- *Name similarity*. Companies that share or closely resemble each other's names through common stems, group prefixes, or near-identical strings are recognized cues of relatedness. It's important to choose a way of measuring "closeness" since a shared relationship is often semantic rather than a literal character match.
- *Shared director/officer.* A common CEO or 代表者 across two separate entities is an instance of an "interlock," or a person connecting two firms. Mizruchi (1996) establishes that such shared individuals are meaningful inter-firm ties (Mizruchi, 1996). In the Japanese context, where the representative director is a registered and publicly available field, this makes 代表者 a directly observable signal.

- *Shared address/co-location*. Research has found that networks built from offshore and registry records repeatedly use shared addresses and shared officers, providing edges that expose otherwise hidden relationships (Joaristi, Serra & Spezzano, 2019; Kejriwal & Dang, 2020). Consequently, two companies registered at the same address can indicate a structural or operational link, and co-location has been a long-standing signal in fraud and anti-money laundering investigations. Moreover, there are journalistic sources whose investigative work also points out the shared address as a signal with reporting of thousands of companies registered to a single address (International Consortium of Investigative Journalists, 2016).

Historically, uncovering these kinds of relationships has been a manual undertaking done by analysts cross-referencing registry entries by hand; early automated systems relied largely on rule-based screening and manually engineered features (Deprez et al., 2026). Furthermore, measurement of similarity between records evolved from hand-built character and token-level string metrics toward learned representations that better capture semantic similarity (Barlaug & Gulla, 2021). On the network side, the past decade has seen a corresponding shift in relationship detection from inflexible rule-based methods toward graph-based and representation-learning approaches (Deprez et al., 2026). These developments are oriented towards improving how investigators measure the similarity between two records, and whether these technologies are capable of catching subtle signals to surface structural associations or will instead mix them with surface-level relationships between entities.

Corporate Vehicles in Japanese Structured Finance

What structures are there in finance that serve a specific purpose and could be detected from universally available fields in the public registry data?

Structured finance is an institutional practice that produces registry signals of relatedness at scale in Japan. One of the main topics in the subject of securitisation which refers to the sale of assets, which generates cash flows and is also employed as part of asset and liability management in order to manage balance sheet risk for financial institutions (Fabozzi, Davis & Choudhry, 2006). In Japanese securitisation law, there are two vehicle types created specifically for these transactions, the specified purpose company (tokutei mokuteki kaisha/TMK) and the specified purpose trust (tokutei mokuteki shintaku/TMS), established under the Act on the Securitisation of Assets (Act No. 105 of 1998) (Nagashima Ohno & Tsunematsu, 2024). Alongside these structures, the market takes advantage of GK-TK scheme, in which a gōdō kaisha (GK) holds the target assets and is financed through tokumei kumiai (TK) contributions

which predominantly is used for real estate securitisation (Nagashima Ohno & Tsunematsu, 2024; Sogo, 2022).

The purpose of these vehicles matter for this study. Firstly, they are single-purpose entities since these securitisation SPVs are typically prohibited from engaging in any business other than the contemplated transaction, so a sponsor executing multiple of them must create a separate vehicle for each (Nagashima Ohno & Tsunematsu, 2024). Secondly, these vehicles made bankruptcy-remote by placing their equity with an independent party (ippan shadan hojin/ISH) established for that sole purpose, with directors drawn from certified accountants or other professionals independent of the sponsor (Chambers and Partners, 2026; Sogo, 2022). Therefore, a GK is routinely accompanied by a paired ISH holding its equity and the professionals who direct these ISHs administer many vehicles side by side. These vehicles are created in batches, one per transaction or asset, and named with a template which makes them detectable in the registry. The Securitisation Act requires a TMK to include special characters in its name and forbids other entities from using them (Act on the Securitization of Assets, 1998, art.16). Structured finance practice inscribes structural relatedness directly into two fields universally available in the public registry, company name and address, which is the layer of the economy a name-and-address representation should be expected to detect effectively.

Similarity Search & Embedding Methods

String matching alone is insufficient for finding related companies when the signal is subtle or distributed across fields. Text embeddings capture semantic similarity that Levenshtein misses, and work on short structured text like company names and addresses.

These registry signals from shared address and representative can be used for surfacing business relationships based on how the similarity between fields is measured. The classical approach to comparing these registry signals is string distance measures, such as Levenshtein distance, Jaro-Winkler, and token-based overlap, and their relative strengths for name-matching are well characterized (Cohen, Ravikumar & Fienberg, 2003). These measures operate by scoring how many characters or tokens the two strings share. This makes them effective against typographic variation but blind to relatedness that is not spelled out in the characters themselves. Since the business associate relational signal could be subtle or distributed across fields rather than concentrated in one, surface matching might miss it.

Learned text representation addresses this by encoding short pieces of text as dense vectors positioned by meaning rather than spelling, so that similarity can be measured as distance in that vector space (Reimers & Gurevych, 2019). A text embedding is a mapping from a piece

of text to a fixed-length vector of real numbers which means that the resulting vector space places texts with similar meaning close by (Vera et al., 2025). Instead of comparing two strings by the characters they share, the embedding model will represent each piece of text such as company name or address as a point in high-dimensional space so the similarity can be measured geometrically by the cosine of the angle between two vectors. Crucially, these positions are not assigned manually, but learned from large text corpora so that text fragments occurring in similar contexts receive similar vectors. The practical consequence is that two pieces of text can be judged similar even when they share few or no characters, provided the implementation context is similar, which is something character-level comparison struggles with. In a task of entity-matching, which is closely-related, replacing hand-built string features with deep and pre-trained language-model representations has been shown to improve performance on short, structured record text (Mudgal et al., 2018; Li et al., 2020).

However, embedding-based similarity carries a specific risk for the task of surfacing business associate relationships. Prior work in relation extraction shows that representations built from entity embeddings tend to capture what the entities are rather than how they are related, producing false positives (Mtumbuka & Schockaert, 2024). If applied in this context, the same signals that reveal a genuine cluster, companies sharing a building and a common name element, can also link firms that merely resemble one another, such as unrelated companies in a densely populated business address. Separating structural association from surface similarity is therefore the central challenge this thesis addresses and the question it tries to answer empirically.

Method

This study begins by creating a single-text representation, or “fingerprint,” for company records in the Kihonjoho CSV. These fingerprints are then embedded into a vector space, where they will be compared against all other records in order to identify pairs of companies with similar representations. Candidate pairs are merged into clusters and then manually reviewed. The pipeline is designed to over-include entities to reduce an intractable comparison space to a small set of reviewable candidates. Nevertheless, the judgment of whether the surfaced relationship is structural or merely superficial, is manual and is the subject of analysis. This design follows the standard entity resolution sequence of standardization, comparison, and classification (Christen, 2012), but the comparison step is replaced with embedding similarity, and the classification step is split between a similarity threshold and human review. The unchanged representative field is carried through to the output and used as a supporting signal during manual review.

Preprocessing and Record Representation

There are ~5.8 million records in the Kihonjoho CSV from gBizINFO distributed across Japan; to make this dataset manageable, this study is restricted to active legal persons whose headquarters' registered address is in Tokyo. This yields 1,173,931 records from the full Kihonjoho CSV. Tokyo contains the densest concentration of shared business addresses in Japan, including virtual office buildings hosting hundreds of registrations and MNCs, which makes it a harder setting for distinguishing structural association from surface co-location and therefore would be the most informative setting for the research question.

Prior to comparison, both company name and address field are standardized, consistent with the standardization stage of the record linkage pipeline (Christen, 2012). The standardization consists of Unicode NFKC normalization, which unifies full-width and half-width character variants, correction of dash-like characters so that the long vowel mark (U+30FC) appears only in kana context while hyphens between digits are rendered as ASCII, removal of whitespace, and case folding. For company names, the decision was made to retain the corporate affixes rather than remove them. Although these affixes denote only the legal form, there is no authoritative list of Japanese legal-form affixes and their variants, so any strip list is necessarily partial. If kept, it could introduce a systematic inconsistency where ordinary joint-stock companies could be treated differently from rarer entity types.

Each record is then represented by a fingerprint which is a single string formed by concatenating the normalized company name and address. Company name (商号または名称) and address (登記住所) are the only two fields with complete coverage in the dataset, so they are the only fields that give every record a comparable representation. The representative field (代表者名称) is excluded from the fingerprint since it is filled for only 1.3% of Tokyo records. It was decided to exclude the representative field that is almost always missing from the fingerprint. On the other hand, including it would make the filled and unfilled records' fingerprints structurally different, possibly pushing potentially related companies apart in the vector space for unrelated reasons.

Embedding

Each fingerprint is embedded with EmbeddingGemma, a 300-million-parameter multilingual text embedding model (Vera et al., 2025), producing one 768-dimensional vector per entity. Embeddings are computed with the model's document prompt and are L2-normalized so that the inner product of two vectors equals their cosine similarity. This gives every pairwise similarity a fixed and interpretable scale, where 1.0 indicates identical direction in the

embedding space, and allows a single global threshold to be applied across all pairs. Character-level measures are effective against typographic variation but blind to relatedness that is not spelled out in shared characters which motivated the choice for an embedding representation instead of classical string distance (Cohen, Ravikumar & Fienberg, 2003). As mentioned before, learned representations place semantically similar short texts close together even when their surface forms differ (Reimers & Gurevych, 2019; Mudgal et al., 2018; Li et al., 2020).

Candidate Pair Extraction

Comparing every record with every other record, even in the Tokyo subset of Kihonjoho, is computationally infeasible, since it would imply 9.3 x 10^11 pairs. This is the blocking problem of entity resolution, where the comparison space must be reduced before the comparison, usually by grouping records on crude keys such as shared leading character (Christen, 2012; Papadakis et al., 2020). This study uses approximate nearest neighbour (ANN) search as a learned form of blocking where instead of a human picked blocking key, the embedding space itself determines which records are comparable.

ANN search is performed with FAISS, a library for efficient similarity search over dense vectors (Johnson, Douze & Jégou, 2019), using a Hierarchical Navigable Small World (HNSW) graph index (Malkov & Yashunin, 2020) with M=32 links per node and efSearch = 64, a standard high-recall configuration for datasets of this size held in memory. Each record queries the index for its k=10 nearest neighbors. The value of k bounds how many relationships per company the search can surface so to keep the output volume manageable. A pair of records is retained as a candidate only if its cosine similarity exceeds a threshold t = 0.9 which acts as the absolute relevance filter. The threshold sensitivity was assessed by repeating the extraction at t=0.85 and t=0.95 on the same embeddings (Table 1). Pair counts at t = 0.90 differ from the main run by two pairs due to the approximate nature of HNSW index construction.

Table 1
Threshold Sensitivity

Threshold (t)	Pairs	Clusters	Largest Cluster Count
0.85	58,212	833,080	296,364
0.90	130,572	39,563	550
0.95	32,860	9,970	372

The output is strongly threshold-dependent in both directions, and the two directions fail differently. Lowering the threshold to 0.85 increases retained pairs sixfold to 833,080, but collapses the candidate graph into a single giant component of 296,364 companies, a quarter of the embedded population. At this point, transitive chaining has percolated and the output is no longer a set of reviewable candidate groups. Raising the threshold to 0.95 reduces the output to 32,860 pairs and 9,970 clusters. The operating value $t = 0.90$ therefore sits in a narrow usable regime between percolation of the candidate graph below it and the loss of the largest structures of interest above it. This narrowness is itself a property of the method worth noting because the threshold window would need to be located empirically for any new dataset.

Clustering

Extracted pairs are merged into clusters by computing connected components with a union-find structure, so any two pairs sharing a member belong to the same cluster. Instead of recovering isolated pairs, it recovers groups; if company A is paired with a company B and B with C, all three form one cluster even though A and C were never directly compared. This behaviour is intended for this study for genuine business associates, but it also means that a single widely shared attribute, such as one virtual office address for many unrelated entities, can chain unrelated companies into one large component. This behavior is not suppressed in the pipeline, because as discussed earlier, representations built from entity attributes capture what entities are rather than how they are related (Mtumbuka & Schockaert, 2024). For each cluster, three descriptive statistics are computed: size, the mean cosine similarity of its internal pairs, and density, defined as the number of retrieved pairs divided by the number of possible pairs among its members. The ranking of clusters is based on mean internal similarity with the size as a tiebreak which will determine the review order.

Evaluation

Since no labelled dataset of business associate relationships exists for Japanese companies, evaluation is a structured manual review relying on human judgment applied to the pipeline's output. The review covered the highest-ranked clusters by mean internal similarity together with the largest clusters, since these two ends of the output represent the method's most confident matches and its most aggregation-prone ones. Rather than imposing a fixed classification scheme in advance, reviewed clusters were characterized individually and recurring patterns were grouped into a typology, which is presented in the Results: duplicate and near-duplicate registrations, genuine corporate groups (predominantly serially named special-purpose vehicles), a professional-administration hub, and surface-coincidence clusters.

Clusters too large for member-by-member review were analyzed structurally, through their address composition, internal pair scores, and density relative to the k-imposed ceiling, rather than classified as a unit.

To test whether the types derived from the edge cases also describe the body of the distribution, a supplementary stratified random sample was reviewed. Clusters were stratified by the size bands and fourteen clusters were drawn with fixed random seed: three pairs, three triples, three clusters of four to nine numbers, three of ten to forty-nine and two of fifty or more. Each sampled cluster was classified against the underlying research question of is it structural association or just a surface resemblance using the same review process. The mapping of types of clusters to binary was fixed before review: duplicates and successions, vehicle families, corporate groups, and co-administered structures count as structural: name-coincidence and address-coincidence clusters count as surface.

Classification draws on the fields carried through to the output for each member, name, address and representative where available, together with manual searching of public sources such as corporate websites and the NTA corporate number lookup. A shared 代表者名称 across cluster members constitutes a director interlock, which the literature establishes as a meaningful inter-firm tie (Mizruchi, 1996), and is treated as supporting evidence of a genuine relationship, with the caveat that an identical name does not guarantee an identical person; conversely, the absence of a shared representative carries little weight, since the field is filled for only 1.3% of Tokyo records. All classifications were made by the author, and the review is subject to the corresponding single-annotator limitation. The composition of the reviewed set across the typology, together with the size and density statistics reported for the full output, constitutes the empirical answer to the question of how much of embedding similarity reflects structural association rather than surface resemblance.

Ethics

All data used in this study is drawn from public sources. As mentioned before, the corporate number, registered name, and registered address of every Japanese legal person are published by the National Tax Agency and re-distributed by gBizINFO with additional fields already held by government bodies. Although personal data such as entity's representative name and title are filled, it is not used as a signal for the embedding, excluded from the fingerprint, and is consulted only during manual review. Moreover, while this research is motivated by the beneficial ownership and anti money-laundering literature which supplied the vocabulary of concealment, corporate vehicles, and due diligence used throughout, it must not be read onto the output. The prominence of some entity's structure in the results reflects how visible they inscribe

their relatedness into the registry, not any property of the entities themselves. The output of the pipeline is a set of candidates for human review, and should not be treated as anything more than what the method measures.

Reproducibility

The repository is structured in folders called data, scripts and notebooks. The main file from gBizINFO (Retrieved 8 July, 2026) should be placed in the data folder, whereas all the exploratory analysis is set in the notebook folder. All processing was implemented in Python using stdlib CSV streaming for data handling, sentence-transformers for embedding, and faiss for indexing. The pipeline consists of two scripts, embed_fingerprints.py and extract_pairs.py. Embedding requires 3.5 hours for 1,173,931 records on consumer hardware (Apple Silicon Laptop, CPU only). The code is available at this link and instructions are provided in the repository README.

Data

The data used in this study is aggregated from two primary sources. First, the National Tax Agency (NTA) publishes the number, registered name, and registered address of every legal person on its Corporate Number Publication Site; this comprises approximately 5.8 million entities which can be downloaded without registration (National Tax Agency, n.d.). The unit of observation is a registered Japanese legal person, identified by its corporate number (法人番号), a 13-digit identifier assigned to every registered legal person under the corporate numbers system introduced in 2015 (National Tax Agency, n.d.). This baseline dataset, however, contains only two substantial fields, besides the corporate number, which is the official name and the headquarters' address.

The second and primary source is gBizINFO, operated by the Ministry of Economy, Trade and Industry (METI). Launched in January 2017, the source aggregates corporate information already held by the government bodies and organizes it under the corporate number, covering roughly over 5.8 million registered legal persons (METI, n.d.). It provides this data through a web search interface, a REST API, and bulk CSV downloads. Crucially for this study, the gBizINFO basic-information table (基本情報) extends the NTA's table with the representative's name and title, postal code, date of establishment and where companies have registered activity information, capital, employee counts and much more.

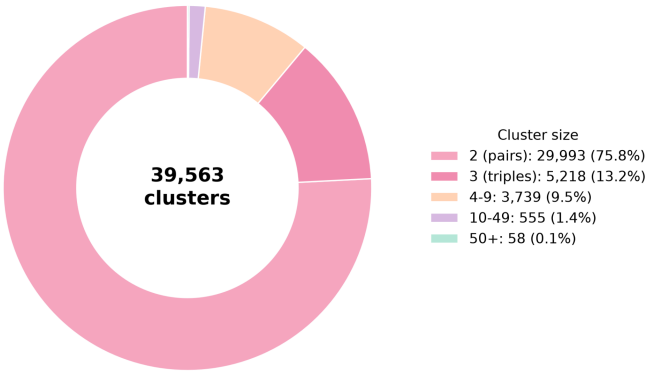
The application is open to both corporate and individual users, but the latter requires contact details and a stated purpose of use, with research and academic use among them, and is subject to acceptance of the service's terms of use (gBizINFO, n.d.). There is no commercial relationship, fee, or institutional affiliation required and the same individual records are accessible without any account through the web search interface. The full study is therefore reproducible by anyone who completes gBizINFO's registration. The dataset used in this thesis is the Kihonjoho_UTF-8_20260708.csv retrieved on 8th July of 2026 through the data download service at info.gbiz.go.jp, containing 5,852,290 records with 96 columns.

Results

The pipeline reduces the full registry to a reviewable candidate set in successive stages. The full Kihonjoho table contains 5,852,290 records, of which 1,365,515 are registered in Tokyo. Excluding records with a closed registry record leaves 1,173,931 active Tokyo entities, which formed the embedding population. ANN self-search at threshold t = 0.9 retains 130,572 candidate pairs, which merge into 39,563 clusters covering 111,833 companies or 9.5% of the embedded population. The remaining 90.5% of companies matched no other records above the threshold and do not appear in any cluster. The retained pair scores concentrate near the threshold: the median similarity is 0.928, the 90th percentile 0.966 and the 99th percentile 0.984. Half of the retained pairs therefore lie within 0.03 of the cutoff, which indicates that the choice of threshold materially shapes the candidate set.

Cluster Size

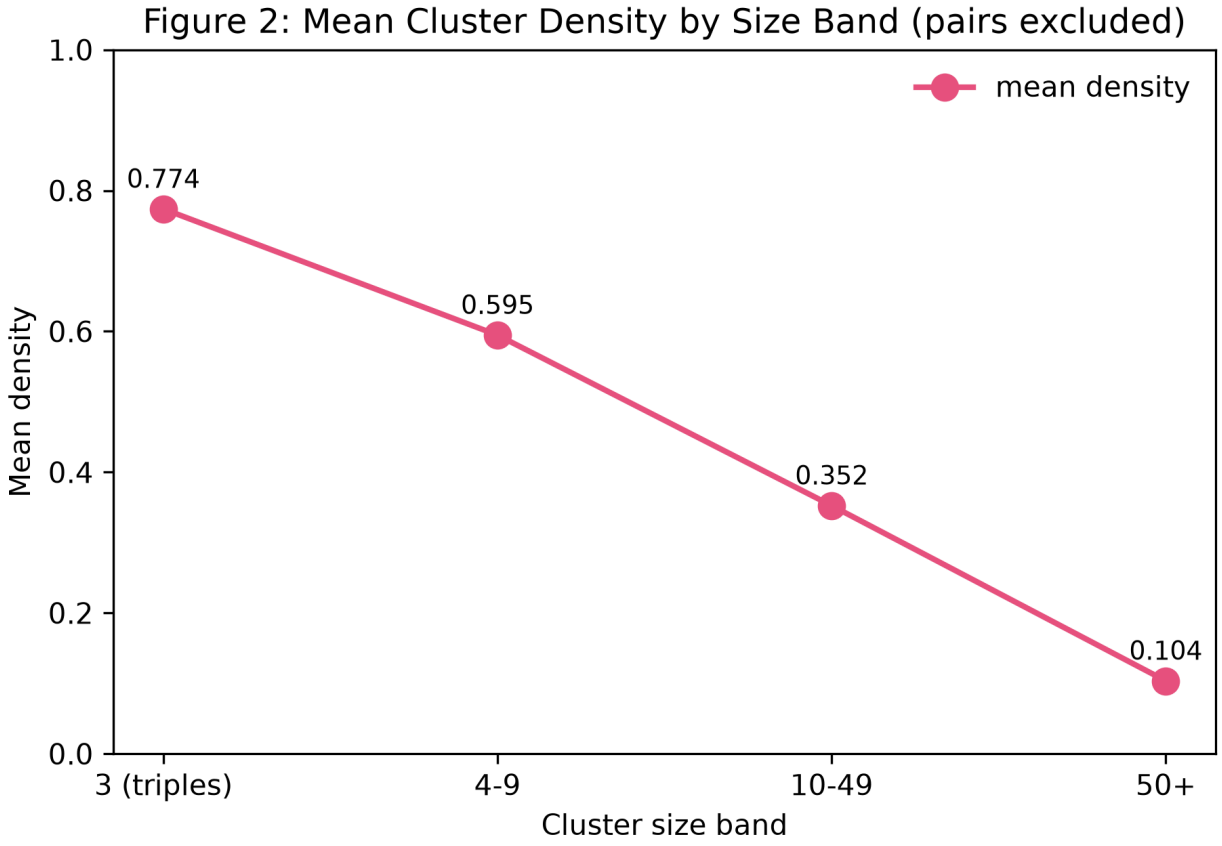
Figure 1: Cluster Size Distribution (t = 0.90)



The cluster population is dominated by small clusters. Of 39,563 clusters, 29,993 (75.8%) are pairs, 5,218 (13.2%) are triples, 3,739 clusters (9.5%) contain four to nine members, 555 (1.4%) contain ten to forty-nine, and 58 clusters (0.1%) contain fifty or more members, the largest containing 550. (Figure 1)

Mean density falls with size: 0.774 for triples, 0.595 for clusters of four to nine, 0.352 for ten to forty-nine, and 0.104 for fifty or more, with pairs at 1.0 by construction (Figure 2). Density comparisons are meaningful within size bands, and for the

largest clusters the informative quantity is retrieved pairs relative to the k-imposed ceiling rather than to the combinatorial maximum, as the case studies below illustrates.



Classification Review

The review of the 10 highest-ranked clusters and the 5 largest, together with clusters flagged during inspection, yielded a typology that structures the remainder of this chapter. Among the highest-ranked clusters, 3 were near-duplicate registration pairs of the kinds analyzed below, and 2 were serial vehicle families. The largest clusters were dominated by the same family pattern, with the eight largest all being serial vehicle families or, in the case of the largest, an administration hub of such vehicles (cluster 8456) as their limiting case, and one large surface-coincidence cluster (21491) as the clearest failure. One reviewed pair (cluster 592) was corroborated by a shared representative name. Moreover, 14 clusters were sampled at random and inspected to reveal structural and surface relationships. The reviewed output is therefore overwhelmingly composed of formulaic structures, duplicates and vehicle families and this

concentration is itself a finding about what name and address similarity surfaces from the registry.

Near-duplicate pairs

The highest-similarity clusters are dominated by duplicate registrations that consist of records with identical normalized name and address under different corporate numbers. Manual inspection shows that these pairs are not a single phenomenon but a spectrum, illustrated by three examples (Table 2).

Table 2
Near-duplicate Pairs Clusters' Information

Panel	Cluster ID	Company Name	Address
(a)	23	株式会社Imagic International	東京都東久留米市滝山7丁目17－24
		株式会社Imagic International	東京都東久留米市滝山7丁目17番24号
(b)	34	株式会社Japan NFT Market	東京都港区海岸3丁目6－32レグノオンザベイ805
		Japan NFT Market株式会社	東京都港区海岸3丁目6－32レグノオンザベイ805
(c)	81	WIT FORTUNE INVESTMENT株式会社	東京都大田区上池台1丁目53番8号
		WIT FORTUNE INVESTMENT有限会社	東京都大田区上池台1丁目53番8号

Panel (a) shows an identical name at an identical location registered once as a domestic joint-stock company (組織種別 301) and once as a foreign company (401), a coexistence of two legal species under one name that may reflect a branch registration alongside a domestic incorporation. Panel (b) shows the same name with the corporate affix in leading and trailing position, registered two months apart at the same address; because affixes were retained in normalization, these records normalize to different strings, and their retrieval demonstrates the case in which embedding similarity recovers a pair that exact matching would miss while preserving the distinction for review. Panel (c) shows an identical name string under two different legal forms, 株式会社 and 有限会社, at one address; since new 有限会社 cannot be formed after 2006, the pair is consistent with a later incorporation by the operator of a

pre-existing entity, that is, with a genuine predecessor-successor or sibling relationship rather than a duplicate.

Beyond these, manual review identified temporal re-registration in which an entity's registry record closes and a successor with the same name and address is registered under a new corporate number. Although the closure-date exclusion removes most of these, but pairs in which both records remain active persist. Duplicates are retained in the output rather than removed, since exact matching on name and address cannot distinguish registry duplicates from legal successions, in which predecessor and successor constitute a genuine relationship; they are instead identified during review and reported as their own category. The prevalence of these cases demonstrates that the corporate number identifies legal persons rather than economic actors, a property of the registry with direct consequences for any entity-level analysis of this data.

The Representative Field as Corroborating Signal

Although the representative field was excluded from the fingerprint due to its 1.3% fill rate, it corroborates classifications during review where present, which cluster 592 illustrates. Its two members, 株式会社オリエンタルコンサルタンツ and 株式会社オリエンタルコンサルタンツホールディングス, share an address in Shibuya and a similarity of 0.9845, and both records carry the same individual as representative, in one record as 代表取締役 and in the other as 代表取締役社長. The pair is an operating company and its holding company where two distinct legal persons that entity-level representation of the registry would render as unconnected nodes, and that the pipeline correctly surfaced as related from name and address alone, with the shared representative confirming the relationship independently. The example also helps to show why the field is usable during review, but unsuitable as a machine signal even when present unless cleaned thoroughly. The representative field includes both the "title", which slightly differs and the field concatenates a varying title with a name whose internal spacing differs across records (野崎　秀則 vs 野　崎　秀　則), so exact matching fails where a human review succeeds (Table 3).

Table 3
A Clusters with a Representative Field

Cluster ID	Similarity	Company Name	Address	Representative Field

592	0.9845	株式会社オリエンタルコンサルタンツ	東京都渋谷区本町3丁目12番1号	代表取締役　野崎　秀則
		株式会社オリエンタルコンサルタンツホールディングス	東京都渋谷区本町3丁目12番1号	代表取締役社長　野　崎　秀　則

Serial Vehicle Families

A recurring pattern among clusters classified as genuine groups is a set of special purpose entries created from a single naming template, distinguished only by a sequence number, and registered at a common address, referred to here as a serial vehicle family. As reviewed in Literature Review, special-purpose vehicles in Japanese securitisation practice are confined to a single contemplated transaction, so a sponsor executing multiple transactions requires a separate vehicle for each (Nagashima Ohno & Tsunematsu, 2024). Therefore, serially numbered entity sets are consistent with this practice; each numbered vehicle in these clusters corresponds to a distinct asset or a transaction. These three clusters illustrate the range.

Cluster 3427 contains twenty-six entities with corporate affix 合同会社 named Resilience I through XXVI, all at one Ginza address, with consecutive blocks of corporate numbers indicating batch registration and registry dates couple of days apart; Cluster 3412 contains thirteen 合同会社桜道 entities numbered between 29 and 54 at a single Minato-ku address, consistent with per-property holding vehicles; the entity numbered 38 was excluded due to it being closed, but other gaps in the sequence are not registered at all; Cluster 3438 contains three ベイエナジー鹿児島 project companies labelled A through C, the one-vehicle-per-site form common in energy projects. These families are the method's clearest true positives because the relatedness is expressed jointly and formulaically in name and address which is retrieved with near-perfect confidence. Their prevalence at scale is visible in the size distribution because the eight largest clusters, together holding 3,043 entities, are all variations of serial vehicle families sharing an exact address and naming template within each group (Table 4).

Table 4
Eight Largest Clusters' Size, ID & Mean Similarity

Cluster ID	Size	Mean Similarity
8456	550	0.9415

7601	421	0.9444
4729	386	0.9565
12363	377	0.9315
3486	370	0.9635
18501	344	0.9211
5909	322	0.9510
3165	273	0.9653

The pattern also delimits what method detects. An operating group whose subsidiaries carry heterogenous names expresses its relatedness in neither field the fingerprint contains, and is correspondingly invisible. Conversely, a group that brands its subsidiaries with a shared stem like any vehicle family shows that visibility of such a layer is a property of the group's naming practice, not its type of structural connection. Therefore, the detection profile of the pipeline is skewed toward the corporate-vehicle layer of the registry.

Case Study: the Largest Cluster

The largest cluster (8456) contains 550 members connected by 2,327 retained pairs. The cluster is not a chain of marginal similarities but a structure in which nearly every member's entire neighbor list lies within the cluster.

Nearly all members are registered at a single Marunouchi address carrying the notation 東京共同会計事務所内 (Tokyo Kyodo Accounting Office), and the membership consists overwhelmingly of special-purpose vehicles with corporate affixes 合同会社, 一般社団法人 and 特定目的会社 entities with serialized names (東京レジ1 through 39, SRFアセット1 through 17, ハウス2 through 53, among others). Many appear in paired 合同会社 and 一般社団法人 forms with identical stems (for example 東京レジ4合同会社 and 東京レジ4一般社団法人). This pairing corresponds to the standard bankruptcy-remote arrangement in which an ippan shadan hojin is established alongside GK (Gōdō Kaisha) to hold all of its equity (Sogo, 2022; Chambers and Partners, 2026), and such pairs are therefore formally related entities. This structure might be due to the nature of Tokyo Kyodo Accounting Office services which are related to accounting, taxation, and finances, focusing on the financial field (Tokyo Kyodo Accounting Office [TKAO], n.d.).

The cluster does not fit a single category of the classification scheme. It combines genuine structural relationships, in the vehicle pairs and serial families it contains, with co-administration ties that fall short of common ownership, since vehicles of different sponsors are administered side by side. It is best described as a professional-administration hub, and it is treated as its own analyzed object. Its detection illustrates both the reach and the limit of the method because the pipeline concentrated the registry's densest visible accumulation of corporate vehicles, the very structures the beneficial-ownership literature identifies as capable of obscuring control (Van der Does de Willebois et al., 2011), while being unable to separate, from name and address alone, the vehicle of one sponsor from those of another.

Case Study: an Instructive Failure

Cluster 21491, the ninth largest with 270 members and a mean internal similarity of 0.9172, is the clearest failure in the reviewed set. Its membership is united by the English word "next" and its variants. The cluster contains distinct companies named simply 株式会社ネクスト or 株式会社NEXT at unrelated addresses across Shibuya, Shinjuku, Minato and beyond, together with dozens of variations on the same aspirational stem: ネクサス, Nexus, ネクステージ, Next Innovation, NEXER, Nexia, Nexight. A common commercial name functions here like a common address by making the fingerprints textually similar without any structural connection between the legal persons. The two mechanisms of the pipeline compounded upon each other to bring this cluster of entities together.

- *Name-coincidence chains.* 株式会社ネクスト matches 株式会社ネクストワン, which matches 株式会社ネクストステージ, and transitive closure fuses them regardless of location.
- *Co-location bridges.* Several addresses recur across many members, notably 西新宿水間ビル (2F and 6F), 渋谷道玄坂東急ビル2F－C, 宮益坂ビルディング609 and 桑野ビル 2階, address patterns consistent with virtual-office or shared-office use. These addresses pull in members whose names have no relation to the stem at all, such as 株式会社High Bridge, 株式会社nano and 合同会社イノウエ, which entered the cluster purely through the address component of the fingerprint and then chained onward.

The result is a single component in which the two surface-similarity mechanisms this thesis distinguishes, shared name element and shared premises, reinforce one another through transitivity.

The cluster's statistics are consistent with this reading: its mean similarity of 0.9172 is barely above the operating threshold and well below the 0.94 to 0.96 range of the genuine serial families and at the bottom of the large-cluster range shown in the table above (Table 4),

indicating that its edges are marginal matches rather than confident ones. Some genuinely related pairs are almost certainly embedded within it, and are rendered unreviewable by absorption into the mass, illustrating that transitive chaining harms not only precision, but also the recall of usable findings. Part of the inflation is addressable by engineering, since richer name and address normalization would separate some variants that currently score as near-identical, but the core failure is representational: the embedding assigns high similarity to shared vocabulary and shared premises alike, without access to the distinction between a naming convention chosen by one sponsor and a common word chosen independently by hundreds of founders. This is the failure mode anticipated in the literature review, in which representations built from entity attributes capture what entities are rather than how they are related (Mtumbuka & Schockaert, 2024), and it is the reason the pipeline's output is treated as candidates rather than findings.

Stratified Random Sample

The clusters reviewed so far were edge cases, the highest-ranked and the largest. To test whether the types produced by the model also describe the middle of the distribution. Fourteen clusters were drawn at random, stratified by size band. Seven were classified as structural, six as surface resemblance, and one as undetermined (Table 5).

Table 5
Fourteen random samples' information

Cluster ID	Band	Similarity	Label	Why
2571	pair	0.9688	structural	GK-ISH pair
12743	pair	0.9307	undetermined	Unverifiable
34577	pair	0.9043	surface	Share only a floor of 有明パークビル
8612	triple	0.9410	structural	Confirmed by a single company profile covering all three
2431	triple	0.9698	structural	Group with an embedded asset vehicle at one address
7266	triple	0.9455	structural	Same company name stem and address
32390	4-9	0.9063	surface	Variants of company name stems (TNS/TKS/TAS), but different addresses
15787	4-9	0.9251	surface	FAM variants
10030	4-9	0.9370	structural	Operating group with the same company

				name and address
11643	10-49	0.9331	structural	TU variants
28439	10-49	0.9100	surface	Variants of common-surname shops (小島屋/小島商店)
21383	10-49	0.9173	surface	Common surname coincidence
12015	50+	0.9322	structural	Letter-coded names at the same address
23062	50+	0.9155	surface	Different names, but same address (virtual-office mass)

The structural cases reproduce the types with two extensions with vehicle layers appearing in the forms. Cluster 2571 is a GK-ISH pair of registered care of an accounting form, with public reporting identifying the GK as the acquisition vehicle for a specific property. Moreover, the cluster 11643 contains eighteen serially coded entities in paired 合同会社 and 一般社団法人 forms, replicating structure of the largest cluster at a smaller scale. This cluster indicates that the professional-administration hub is a recurring form in the registry rather than a property of a single office. Cluster 12015 is a family of 82 ジェイ-series entities in 株式会社, 合同会社 and 一般社団法人 forms at one Shinjuku address, serialized by trailing letter combinations, which shows that the naming template can use letters, instead of a numeric sequence. Cluster 7266 pairs two numbered エムシーエフ・プランニング 有限会社 with a 有限責任中間法人 of the same stem at one Ginza address.

The remaining structural cases are ordinary operating groups rather than vehicle structures. Cluster 8612 consists of three 八正 construction companies in Nerima is confirmed by a single corporate profile covering all three members. Cluster 2431 contains the Accordia Golf holding company, operating company, and an-asset holding entity at the same address. Moreover, cluster 10030 groups seven 株式会社オーガスタ entities, including a ホールディングス, in one Shibuya building. These retrievals qualify that a group is visible exactly to the extent that it brands its subsidiaries with a shared stem and registers them together, which seems to be a common appearance.

The surface cases likewise reproduce the two failure mechanisms at different scales. Things like name coincidences (cluster 15787), unrelated initialisms (cluster 32390), and similar family names (cluster 28439 and 21383) which seem to be the reasons for structurally unrelated companies to appear in the same cluster. Additionally, address similarity appears in cluster 34577, members being a SRE engineering firm and an unrelated corporate group, that share a

floor of one Ariake building with a similarity of 0.9043, the weakest edge in the sample. Both of these mechanisms appear combined in cluster 23062, a 207-member cluster assembled around the shared office suite in Shibuya with several recurring addresses. The two largest surface-coincidence clusters in the output are generated in part by the same physical virtual-office infrastructure, the surface counterpart of the administration hubs on the structural side. The 小島 and 鈴木 clusters contain records whose addresses remain in the ward notation of Tokyo City (東京市) which was abolished in 1943, later merging into the newly created Tokyo Metropolis (Tokyo Metropolitan Government, 2026). These records predate the war and received corporate numbers when the system assigned an identifier to every existing legal person at its introduction in 2015 (National Tax Agency, n.d.).

One pair, cluster 12743, was classified as undetermined due to the idea that co-location and the shared name element make a family or ownership relationship plausible, but no public source confirms or refutes it. These kinds of cases illustrate the verification ceiling of the evaluation itself for entities with no public presence beyond the registry. Lastly, the internal similarity of the sample separates the two classes: the six surface clusters score between 0.9043 and 0.9251, the seven structural clusters between 0.9322 and 0.9698, so the single undetermined pair falls inside the gap at 0.9307. A sample of fourteen cannot establish a decision boundary for clusters' classification, and the bands overlap in the full output stays untested. Nevertheless, there is an observed contrast between the serial families (0.94-0.96) and the failure cluster (0.9172), so within this sample, mean internal similarity would have scored the structural associations higher than the surface ones. The random sample as a whole establishes that the types extracted from the extremes did not require new categories of classification for randomly drawn clusters; administration-hub/virtual-office patterns each reappear at different addresses; and the association between marginal similarity and surface resemblance holds. This sample's size supports these conclusions but not the general statistical estimates of the type's composition in the full output.

Discussion

This thesis is centered on finding out how much of embedding similarity in public registry data reflects structural association between companies and how much reflects surface resemblance. The method reliably surfaces relatedness that turned out to be formulaic which is multiple registrations of one entity, serially named special-purpose vehicles, paired GK and ISH structures, and entities gathered under a common administrator. What embedding similarity over name and address detects is not "business association", but the subset of association that entities inscribe into their registrations.

The results give empirical, registry-scale form to a concern raised in the relation-extraction literature, that representations built from entity attributes capture what entities are rather than how they are related (Mtumbuka & Schockaert, 2024). Each of the main findings is an instance of this single phenomenon. The failure cluster formed because hundreds of unrelated individuals independently chose the same word as a base for a company name in business dense areas of Tokyo like Shibuya, Shinjuku, and Minato. The administration hub, cluster 8456, forms because a shared administrator address makes vehicles of different sponsors textually near-identical, and the embedding cannot separate co-administration from common control. Nevertheless, the method retrieves the serial vehicle families almost perfectly precisely because their relatedness is written into their attributes by construction.

The threshold sensitivity analysis extends this from individual clusters to the behavior of the whole candidate graph. Between $t = 0.85$ and $t = 0.95$, the output passes from a giant component covering a quarter of the population to the dissolution of the largest structures of interest. The usable operating threshold is narrow and its location is a property of the dataset instead of the method, so it should be found empirically for any new corpora. Taken together with the transitivity failures, this supports that embedding-based blocking over registry attributes is a candidate generator whose output requires human classification, not a relationship detector. The design of the pipeline used in this study over-includes and defers judgment to review is the form this class of method took, instead of being merely a practical choice.

The stratified sample supports extending these conclusions beyond the reviewed extremes. Within randomly drawn clusters, the same administration hubs and virtual-office mechanisms recurred through the same bridge addresses as in large clusters.The failure and the success of the method are not properties of individual clusters, but two recurring kinds of relationships, structural and surface, that the embedding cannot tell apart. The method performs best on the entity type that the beneficial-ownership literature previously is most concerned with. The corporate vehicles that van der Does de Willebois et al. (2011) identify as the near-universal instrument of ownership concealment follow the same structure as found in the method. They are confined to single purposes, registered in batches, named from templates, and administered side-by-side. These are the vehicles of Japanese structured finance, TMK and TMS, and the GK-TK scheme with its paired ISH whose single purpose, batch-registered, template-named characteristics is precisely why they are visible to this study's name-and-address representation. The detection profile was heavily matched to this kind of corporate-vehicle layer of the registry and poorly matched to the ordinary operating economy. It was visible directly in the Tokyo subset of the Kihonjoho file because the eight largest clusters, holding 3,043 entities, are all serial vehicle families, and the single largest is an administration hub of securitisation vehicles.

The second contribution of the results is a measured account of what Japan's open corporate data can and cannot support, which was previously mentioned as the gap between FATF compliance on paper and access in practice. The measurements define the ceiling:

- The representative field. The only representative information in the dataset is filled for 1.3% of Tokyo record, and cluster 592 shows that even where present it can resist the machine use unless standardized. Even if the representative field is present in the cluster, there is no way of proving if it's the same person due to the nature of the Japanese names.
- The lack of shareholder field. The Kihonjoho dataset contains no shareholding or ownership fields at all, so ownership-based resolution is undefined and difficult on this data.
- The corporate number. The 法人番号 (corporate number) identifies legal persons rather than economic actors so the near-duplicate spectrum shows one economic entity appearing under multiple rows through concurrent filling, re-registration, and succession. Some clusters have entities with 東京市 notation in their address (pre-1943) which received the corporate number at the system's introduction in 2015 and remains active.

The consequence of this is that public registry data supports associate identification only for relationships that entities choose to broadcast through naming and co-location. R24, as revised, required that competent authorities have access to beneficial ownership information, but does not extend that access to the public which this study provided a measurement for. The proxy signals available in public data highlight the visible part of the corporate-vehicle layer, and the rest is about finding the data.

Implications

This method worked to produce a reviewable set of candidates with strongest structures, families and hubs from a 1.17 million entity registry. It can not become a substitute for the agencies and commercial investigators whose mandate is surfacing fraudulent structures beyond missing fields. The corporate number system itself dates only to 2015, and numbers were assigned to all the existing at the time legal persons, so for the large population of legacy companies the dataset's temporal fields mark the arrival of an identifier rather than company's history. The layers structure of ownership and control chains constructed before 2015 is not observable through this data. Therefore, the pipeline's role is confined to the present-day surface of the registry that can help compliance officers by providing a first pass that concentrates candidates for human review. Within that role, given research and development time, the patterns can underlie a service in identity resolution and risk assessment. It can provide a continuously

updated candidate layer over the public registry, flagging vehicle families, administration hubs, and re-registration events as they appear. To unlock more analytical value for such services, it would be best to use data that also holds historic updates such as re-registration, name or address change, and has a highly populated, standardized representative field.

Limitations

The limitations are grouped into three kinds. In the design scope, the study covers a single prefecture at a single date (8th July of 2026), uses one embedding model, and fixes k at 10. The sensitivity was assessed for the threshold ($t = 0.9$), but not for k, whose value caps the measurable density of large clusters and may undercount within-family pairs for families larger than ten. In the evaluation scope, all classifications were made by a single annotator with no ground-truth labels to check against and the main review concentrated on the highest-ranked and largest clusters, supplemented by a stratified random sample of fourteen clusters whose size supports qualitative confirmation of the typology, but not statistical estimates of its composition. In the method ceiling, part of inflation of the cluster 21491 is addressable by richer normalization of names and addresses, but the core limitation is that a name-and-address embedding cannot distinguish structural association from surface resemblance because the distinction is not present in the text it reads.

A final limitation of this study is the shift of the framing. The study was designed around beneficial-ownership concealment, and the prominence of securitisation structures in the results such as serial vehicle families and administration hubs was not anticipated at design time. Asset-holding practice where sponsors routinely letter and number of single-purpose entities was encountered during analysis rather than surveyed beforehand, which is noted here in the interest of transparency.

Future Work

Each limitation on its own indicates a direction for improvement. Even though the contribution of this improvement to the position of the entity in the vector space is not thoroughly explored, a better normalization pipeline for addresses that correct kanji numerals, 丁目-番-号 notation, and regional conventions such as Kyoto's directional addresses to bring it to one standardized form could help bring cleaner results. If textually different rendering of one address embed far apart, richer canonicalization would directly improve recall, but if the model

maps the addresses close, effort is best spent elsewhere. The fingerprint also carries tokens shared by every record in the corpus, the prefecture name and corporate affix, which contribute a constant component to every pairwise similarity and compress the usable score range. Beyond preprocessing, per-field embeddings with score-level fusion, in place of the concatenated fingerprint, would restore attribution of which signal drove each match, at the cost of a fusion weighting to calibrate. Moreover, the embedding model, EmbeddingGemma, supports fine-tuning through the sentence transformers framework which could be implemented for this specific task to better score clusters. Finally, the evaluation itself could be grounded by obtaining representative or shareholder data for a subset of entities to help measure the recovery of known relationships and resolve the registry inconsistencies mentioned before.

Conclusion

This thesis built and evaluated a pipeline that surfaces candidate business-associate relationships from Japan's public corporate registry using fingerprint embeddings and approximate nearest neighbour search, applied to 1.17 million active Tokyo entities. Its central finding describes an asymmetry: relatedness that is inscribed formulaically into names and addresses is retrieved with high confidence, while the method's errors arise from surface patterns that imitate that formula through common words and shared premises. In the process, the study measured the practical ceiling of Japan's open registry data for network analysis with limited access to representative's name, no shareholder information and the lack of an entity identifier that does not survive re-registration. Within that ceiling, the method demonstrates that even two universally available fields read semantically can showcase the corporate-vehicle layer of the economy that the beneficial-ownership literature identifies as the primary instrument of concealment.

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